# torch.frexp#

https://pytorch.org/docs/stable/generated/torch.frexp.html

**Description:**

Decomposes input into mantissa and exponent tensors such that input=mantissa×2exponent\text{input} = \text{mantissa} \times 2^{\text{exponent}}input=mantissa×2exponent. The range of mantissa is the open interval (-1, 1). Supports float inputs. input (Tensor) – the input tensor out (tuple, optional) – the output tensors

## Function Signature

```
torch.frexp(input, *, out=None) -> (Tensor mantissa, Tensor exponent)#

Parameters

Keyword Arguments
```

## Parameters

### Keyword

out (tuple, optional) – the output tensors

## Code Examples

```
>>> x = torch.arange(9.)
>>> mantissa, exponent = torch.frexp(x)
>>> mantissa
tensor([0.0000, 0.5000, 0.5000, 0.7500, 0.5000, 0.6250, 0.7500,
0.8750, 0.5000])
>>> exponent
tensor([0, 1, 2, 2, 3, 3, 3, 3, 4], dtype=torch.int32)
>>> torch.ldexp(mantissa, exponent)
tensor([0., 1., 2., 3., 4., 5., 6., 7., 8.])
```

# Detailed Documentation

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